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# Fruit Logistics

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**Tesse Bouwmeester, Yunji Eo, Derek Nigten,  
Eveline Rietveld**

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Sean Telg and Arantza Estévez-Fernández

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### Abstract

This paper aims to provide guidance to the three supermarket chains in whether to accept the service of a logistic company, Fruit Logistics (FL), or individually handling transportation and storage. To achieve this, OLS estimation is used for the daily demand and the total costs and cost savings are determined. Subsequently, an LP program is used to solve the problem. Game theory is used in the decision to accept the service.

**Keywords:** logistic company, daily demand, OLS estimation, total costs, cost savings, LP program, game theory

## Introduction

In this report, Fruit Logistics (FL), a company specialized in fruit transportation, faces a challenge in optimizing the transportation and storage of bananas for three major supermarket chains: Albert Hijné (AH), Junco (J) and Koop (K). In order to ensure that the supermarket chains choose FL to transport their bananas instead of handling the transportation themselves, FL needs a convincing plan. The proposed deal involves sharing a portion of the cost savings. There is an optimal approach in what order to approach the supermarket chains, in the expectation that they will accept the services of FL.

The study of this topic is essential since it is an representative case of modern supply chain management, namely the challenges that logistics companies face when ensuring cost-efficient and timely delivery of goods. This will not only impact the profitability of the involved companies, but also influences consumers in the availability of essential food products.

Previous research in logistics for supply chain networks has mostly focused on general optimization models. However, the challenge for FL nuances on the transportation of specific goods, such as bananas, in combination with multiple chains. This needs a more specific approach than general optimization. There are some examples of findings in logistics in a specific field. Recently, research has been done in the specific field E-commerce, where worldwide implementations and corresponding models together with supporting techniques are reviewed (Yu et al., 2016).

The objective of this study is to optimize the transportation and storage of bananas for each supermarket chain through collaboration, being facilitated by Fruit Logistics. Thereby, the paper contributes to the existing literature since it combines finding transportation strategies by linear programming and game theory. The hypothesis underlying this study is that costs can be reduced for all supermarket chains by working together in transportation. Specifically, we hypothesize that collaboration leads to a substantial decrease in transportation costs for each supermarket chain compared to individually handling transportation and storage. The idea behind this hypothesis is that by strategically determining the optimal order of approaching the chains, routes are optimized and resources will be shared. This strategic advantage not only increases collaboration with supermarkets but also makes banana transportation overall more efficient.

To determine the optimal order in which FL should approach the supermarkets and earn the maximum cost savings, it is essential to establish an optimal route for transportation and storage costs for the bananas. However, to do so, a thorough analysis of the available data must be conducted first. Here, different variables need to be linked to the data. Besides, several models and techniques are needed to analyze the data, which are explained in the methodology. In the data analysis the findings of the optimal cooperation between the supermarket chains are presented. The implications of the results will be analyzed in the discussion and the key findings will be summarized in the conclusion.

## Data

The data used in this study has been extracted from an excel file: "logistics4.xlsx", containing twelve different sheets. Each sheet provides unique information into the logistics of Fruit Logistics and the supermarket chains Albert Hijne, Junco and Koop. The data covers diverse aspects such as transportation costs of each chain, unloading times, the time and distance between location and product details. The first sheet labeled as 'costs' shows the transportation costs of every chain and of Fruit Logistics, with important parameters such as fixed costs and storage costs. The next six sheets are labeled as days Monday till Saturday, which show 222 observations in square meters, customers/day and bananas/day. The number of observations and the bananas per day are important when determining the daily demand for each day of the week. Sheet eight and nine provide information on the travel time and travel distances between different cities. In addition, sheet ten, eleven and twelve outline the sizes of supermarkets in different cities for the supermarket chains Junco, Albert Hijne and Koop respectively.

The variables are interconnected with the data and form crucial parameters. For example, it is clear to see that the square meters of the store are linked to the bananas per day. The square meters (data) are a measure of the size of a store (variable) and the bananas per day (data) are a measure of the demand of bananas (variable). Furthermore, minutes (data) from city to city are used as a measure of travel time (variable) and kilometers (data) from city to city are used as a measure of travel distance (variable).

## Methodology

Now, we will first explain a few methods because we need some additional background information for the further analysis for FL.

### Linear regression model

If one variable  $X$  relates to another variable  $Y$ , the linear relationship can be postulated by a linear regression model. Such a relationship can be for example the size of a store and demand of bananas, which are variables appointed in Data. Then, the intercept and slope of the line relating  $X$  and  $Y$  are unknown characteristics of the population joint distribution of  $X$  and  $Y$ . The econometric problem is to estimate the intercept and slope using a sample of data on these two variables (Stock and Watson, 2020). When  $X$  can take on multiple values, the easiest starting point for modeling a function of  $X$  is to suppose it is linear. This gives the linear function:

$$Y_i = \beta_0 + \beta_1 * X_i + u_i \quad (1)$$

This equation is called the linear regression model with a single regressor, where  $Y$  is the dependent variable and  $X$  is the independent variable or regressor. The first part of the equation,  $\beta_0 + \beta_1 * X_i$ , is the population regression line. Given the value of  $X$  the value of the dependent variable  $Y$  can be predicted.  $\beta_0$  is the intercept of the population regression line, which is the point at which the line intersects the  $Y$  axis.  $\beta_1$  the slope of the population regression line, which shows the difference in  $Y$  associated with a unit difference in  $X$  (Stock and Watson, 2020). When showing the correlation between the two variables  $X$  and  $Y$ , a scatterplot can be made (Rensink and Baldrige, 2010). This shows the population regression line  $\beta_0 + \beta_1 * X_i$  and several points. The vertical distance from point  $i$  to the population regression line is the population error term  $u_i$  for the  $i^{th}$  observation.

Sometimes, several variables are omitted which can lead to a misleading estimate. Multiple regression includes omitted variables as additional regressors and can estimate the causal effect of one regressor while holding constant other variables. The multiple regression model is given by:

$$Y_i = \beta_0 + \beta_1 * X_1 + \dots + \beta_p * X_p + \epsilon \quad (2)$$

Note that equation (2) becomes the single regression model (1) if  $p = 0$ , which is the model we need in this study. Hence, we only explain OLS for the single linear regression model.

To examine how well the regression line approximates the real data points, a measurement of the fit  $R^2$  can be used. This is the fraction of the sample variance of dependent variable  $Y_i$  explained by the independent variable  $X_i$ . The more  $R^2$  gets to the value 1, the more perfectly the regression model fits the data.

## Ordinary Least Squares

Ordinary Least Squares (OLS) is a method to estimate unknown parameters in the linear regression model, as described above (Stock and Watson, 2020). The estimators of the unknown true population intercept and slope that minimize the sum of squared mistakes are the ordinary least squares estimators of  $\beta_0$  and  $\beta_1$ . The OLS estimator of  $\beta_0$  is  $\widehat{\beta}_0$  and the OLS estimator of  $\beta_1$  is  $\widehat{\beta}_1$ . The OLS regression line is constructed using the OLS estimators:  $\widehat{\beta}_0 + \widehat{\beta}_1 * X$  (Stock and Watson, 2020). The predicted value of  $Y_i$  given  $X_i$  can be calculated by the following equation:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 * X_i \quad (3)$$

The OLS estimators  $\widehat{\beta}_0$  and  $\widehat{\beta}_1$  could be found by trying different values of  $b_0$  and  $b_1$  multiple times until the values are found that minimize the total squared mistakes. There is a formula to calculate the OLS estimators of the slope  $\beta_1$  and intercept  $\beta_0$ :

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad (4)$$

$$\widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 * \bar{X} \quad (5)$$

In these equations the estimated intercept, slope and residuals are computed from a sample of  $n$  observations (Stock and Watson, 2020).  $\bar{X}$  and  $\bar{Y}$  are the sample average. Now it is possible to predict the dependent variable  $Y_i$  given a value of the independent variable  $X_i$  by using OLS.

We use this method since it is the common language for regression analysis and because OLS is easy to use. The OLS estimator is theoretically efficient among other unbiased estimators.

There are three mathematical assumptions under which OLS estimates the causal effect between the two variables  $Y$  and  $X$ . These three assumptions make sure that the OLS estimator is unbiased and consistent.

- Assumption 1: The conditional distribution of  $u_i$  given  $X_i$  has a mean of zero.

The error term  $u_i$  represents the contribution to the dependent variable  $Y_i$  made by all other omitted factors than  $X_i$ . This assumption states that all other factors contained in  $u_i$  are unrelated to  $X_i$ , also known as strict exogeneity.

- Assumption 2:  $(X_i, Y_i), i = 1, \dots, n$  are independently and identically distributed.

This assumption is a statement on how the sample is drawn and provides that the sampling distributions of the OLS estimators can be approximated by a normal distribution in large samples. Both variables come from the same population and are randomly selected.

- Assumption 3: Large outliers are unlikely.

OLS regression results can be misleading due to large outliers. So, this assumption is made mathematically precise by assuming that  $X$  and  $Y$  have nonzero finite fourth moments:  $0 < E(X_i^4) < \infty$  and  $0 < E(Y_i^4) < \infty$ .

There are also two extended least square assumptions, which can provide some extra information on the reliability of our results.

- Assumption 4: The error term has a constant variance and in particular does not depend on the independent variable.

This means the variance does not change for each observation or for a range of observations. This assumption is known as homoskedasticity (same scatter).

- Assumption 5: The conditional distribution of  $u_i$  given  $X_i$  is normal.

In other words, this means that the error term is normally distributed. When this assumption is satisfied hypothesis testing may be performed and confidence and prediction intervals can be generated.

When assumption 4 is violated, we say that the error term  $u_i$  is heteroskedastic, defined as the unequal scattering of residuals. If this occurs, the results of the regression become unreliable. Therefore, this assumption needs to be tested, which can be done by the Breusch-Pagan Test. By regressing the squared residuals on the independent variables, it tests if the variance of the error term of  $u_i$  depends on a certain variable  $z_i$ :

$$\hat{u}_i^2 = \alpha_0 + \alpha_1 * z_i + \epsilon_i \quad (6)$$

Where we test:

$$\begin{aligned} H_0 : \alpha_1 &= 0 (\text{homoskedasticity}) \\ H_1 : \alpha_1 &\neq 0 (\text{heteroskedasticity}) \end{aligned} \quad (7)$$

The null hypothesis can be rejected if the p-value is lower than a chosen significance level, mostly 5 percent. This shows evidence that the error term is heteroskedastic.

After testing for homoskedasticity, normality can be tested (assumption 5). This order is important, because testing the other way around leads to unreliable results. This can be done by the Jarque-Bera Test, a goodness-of-fit test that determines whether or not sample data have skewness and kurtosis which matches a normal distribution. The skewness is a measure of the asymmetry of the probability distribution of  $X$  and the kurtosis is a measure of the "tailedness" of the probability distribution of  $X$  (Stock and Watson, 2020).

$$\begin{aligned} H_0 : \text{skewness} &= 0 \text{ and } \text{kurtosis} = 3 \\ H_1 : \text{skewness} &\neq 0 \text{ and/or } \text{kurtosis} \neq 3. \end{aligned} \quad (8)$$

The null hypothesis can be rejected if the p-value of this test is lower than the chosen significance level. This shows the data has skewness and kurtosis that is significantly different from a normal distribution.

If all five assumptions hold, we can say that the OLS estimator is unbiased, consistent, asymptotically normal and efficient with a large sample size (Stock and Watson, 2020).

When heteroskedasticity exists, another method can be applied to fix it. This method is Weighted Least Squares, which places weights on the observations. The observations with small error variance will get more weight since those contain more information in comparison to observations with larger error variance.

## Linear Programming

Linear programming is a technique to solve Operations Research problems in practice. This method is commonly used since it provides a well-defined mathematical framework. A general constrained optimization problem involves finding the maximum or minimum of a function subject to a set of constraints (Bertsimas and Tsitsiklis, 1997). A linear program is a special case of the general constrained optimization problem with a linear objective function and linear constraints. A linear program has the following form:

$$\begin{aligned} \max & f(x) \\ g_i(x) & \leq b, i = 1, \dots, m \\ \text{with } & f, g : \mathbb{R}^n \rightarrow \mathbb{R} \end{aligned} \quad (9)$$

Where a linear program has several properties:

1. Separability: Each function is the sum of terms, in which each term is a function of one of the variables only.

2. Proportionality: The value of the function is proportional to the value of each of the variables.
3. Divisibility: All variables may take any values in  $\mathbb{R}$ .

Modelling with a Linear Program consists of several steps. Firstly, the decision variables need to be chosen. These variables are usually denoted by  $x$ ,  $y$ ,  $z$ . After choosing the decision variables, the objective function needs to be chosen. This is a linear function of the form  $\sum_{j=1}^n c_j x_j$ . The letter  $c$  is usually used for the so called objective coefficients (Bertsimas and Tsitsiklis, 1997). The objective function is to be maximized or minimized, denoted by respectively  $\max$  or  $\min$  in front of the function, depending on the problem. Lastly, the restrictions are constructed. Restrictions can be equalities as well as in- equalities. A functional restriction or constraint is of the form  $\sum_{j=1}^n a_{ij} x_j \leq b_i$ . If variables are required to be non-negative, non-negativity constraints  $x_j \geq 0$  need to be added. Feasible solutions are sets of values assigned to decision variables that satisfy each constraint. The set of all such feasible solutions is called the feasible region. If the feasible region contains at least one solution, the problem is considered feasible.

## Vehicle Routing Problem

The Vehicle Routing Problem (VRP) is one of the most important combinatorial optimization problems in the field of operations research and logistics. The main objective of the VRP is to determine the optimal set of routes to deliver goods or provide services to a set of given customers performed by a fleet of vehicles, while satisfying various constraints and minimizing overall costs (Toth and Vigo, 2002). Vehicles despatched from a single location must deliver the goods required and then return to the location. This vehicle is limited in the weight it can carry and can also be restricted in the total distance it can travel (Baker and Ayeche, 2003).

There are several VRP variants due to different factors influencing the problems, such as time-related restrictions and the existence of multiple locations. The objective stays the same in most cases which is the minimization of the total distribution costs (Konstantakopoulos et al., 2020). We analyze the variants that most logistics and distribution companies face in their daily operations.

- The capacity of vehicles: Each vehicle has a limited capacity where the total demand delivered by a vehicle can not exceed its capacity, known as the capacitated VRP (CVRP). In the Multi-Trip VRP vehicles execute more than one trip during the working day.
- Requirement of the customers: Each customer determines a time interval in which the order must be delivered, this is called a VRP with time windows (VRPTW).
- Multiple distribution centers: If a company manages multiple distribution centers, customers are most commonly assigned to the nearest warehouse, known as the Multi-Depot VRP (MDVRP).

Algorithms are used to commonly deal with the containing variants in order to simplify the problem (Konstantakopoulos et al., 2020). Metaheuristic algorithms attempt to find the best feasible solution out of all possible solutions of an optimization problem, which is used to find the most optimal routes in the VRP. A high-performance implementation of the Hybrid Genetic Search (HGS) algorithm, specifically designed for solving VRPs can be utilized to address a CVRP (pyV, nd). The algorithm proceeds through a sequential process, starting with the initialization phase wherein it generates a list of solutions. Once this list is available, the algorithm proceeds to select two parents, recombining their features to generate a new solution that incorporates characteristics from both parents. Following that, the algorithm refines the solution using a local search, and the resulting outcome is inserted back into the population. This iterative procedure is repeated until a criterion is met (Vidal, 2022).

For some of the variations of the VRP linear programming formulations can be derived to solve the problem, to model it as in (9). To do this, some aspects need to be linearized to make it amenable to linear programming techniques.

## Game Theory

Game theory is a powerful method for modeling and analyzing strategic interactions among decision-makers. The game theoretic model of interdependent choice refers to a situation in decision-making where the choices or decisions of individuals or entities are influenced by the decisions of others (Houba, 2023). When agents act independently and make decisions without the ability to form binding agreements, it is called Non-Cooperative Game Theory. In these games, game theory helps identifying optimal strategies for individuals when looking at their own interests. Individuals can also choose to form coalitions, make binding agreements, and cooperate to achieve joint outcomes, called Cooperative Game Theory (Houba, 2023). Moreover, players can also take decisions simultaneously without having knowledge of others or players can take decisions sequentially where they can observe the actions of those who have already made decisions.

### Cooperative Game theory

Cooperative games are strategic scenarios where players have the opportunity to negotiate and form a group, known as coalitions, aiming to achieve shared objectives. To model such games, the concept of a characteristic function form, represented as the pair  $(N, v)$ , is crucial. Here,  $N = \{1, 2, \dots, n\}$  signifies a finite set of players participating in the game. The function  $v$  assigns a total payoff to each subset of  $N$ , with the value of the empty set  $v(\emptyset)$  is zero (Michael Maschler, 2020).

The Shapley value is a method for the fairly distribution of the game's total payoff among its players. It achieves this by considering all possible sequences denoted as  $\pi$  in which players can form the coalition. For each sequence, the method calculates the individual contribution of a player based on the difference made by their joining the coalition. The Shapley value is then the average of these contributions across all possible permutations of player arrangements. The Shapley value for every permutation  $\pi \in \Pi(N)$  is expressed as (Michael Maschler, 2020) :

$$Sh_i(N; v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} (v(P_i(\pi) \cup \{i\}) - v(P_i(\pi))), \quad \forall i \in N. \quad (10)$$

### Games in normal form

The model to study interdependent decision making with simultaneous actions is called a game in normal form. A game in normal form is defined by the tuple  $\langle N, A, u \rangle$ , with  $N$  the set of players (decision makers),  $A$  the set of action profiles and  $u$  the utility function. The best action is determined by utility maximization over the set of feasible actions of a player (Houba, 2023). A Nash Equilibrium is a prediction robust against unilateral deviations. To find a Nash Equilibrium, a graphical method can be used. Every players needs to underline their best response given the other player's action. A Nash Equilibrium is Pareto efficient if no player can be made better off without making another player worse off. There are critical assumptions underlying games in normal form:

1. Consistent predictions
2. Coordination of Nash equilibrium predictions
3. Self-fulfilling predictions
4. Flawless choices of best responses

### Games in extensive form

Analyzing games where players move sequentially is done with games in extensive form. To represent sequential decision-making and the dynamic nature of strategic interactions, a game tree structure is used. Gambit is freeware where nice figures of games can be represented (Savani and Turocy, 2024). The tree consists of nodes assigned to players, which are called decision nodes (Houba, 2023). If all players have complete information about the game and the actions taken

by previous players, it is called perfect information. With imperfect information, players have incomplete knowledge about aspects of the game. Some players have private information. If all information sets consist of a single decision node, there is perfect information. Multiple decision nodes in the information set shows imperfect information. Backward Induction Equilibrium is defined as a solution found by solving the extensive form backward from the end to beginning, which can be used with perfect information. There are several assumptions underlying backward induction equilibrium:

1. Consistent predictions
2. Coordination of equilibrium predictions
3. Self-fulfilling predictions
4. Flawless implementation of equilibrium strategies
5. Cognitive abilities

If there is imperfect information, smaller extensive forms in the entire game need to be identified, called proper subgames. A proper subgame consists of a single decision node. Subgame perfect equilibrium extends the idea of backward induction equilibrium to sequentially solving Nash equilibrium backward, starting with the smallest proper subgame at the end (Houba, 2023). This ensures that the strategies of the players are optimal not only in the overall game but also in every subgame. The assumptions are the same as above.

## Data analysis

### Daily demand

Firstly, the FL headquarters need to determine the bananas transportation needs of each supermarket chain: Albert Heijn, Jumbo and Koop. To estimate this daily demand of bananas per day per supermarket chain Ordinary Least Squares (OLS) estimation is used, explained in the methodology. In Data we saw that the collected data relating to the square meters are a measure of the size of the store and that the bananas per day are a measure of the demand of bananas. Since the size of the supermarket store affects the demand of bananas, the relationship of these two variables can be shown in a Linear Regression Model. When using equation (1) from methodology, the demand of bananas is the dependent variable  $Y_i$  and the size of the supermarket store is the independent variable  $X_i$ . If we use the demand of bananas as  $Y$  and the supermarket store size as  $X$ , a scatterplot can be made to get information about how the data is distributed. This gives a good overview to see the correlation between the two variables. For every day separate, the data points are plotted and the regression line  $\beta_0 + \beta_1 * X_i$  is added to check for any outliers. Since this gives many plots we chose three days (Monday, Wednesday and Saturday), this can be seen in Figure (1). The remainder can be observed in the appendix. In the plots the  $X$  variable store size is placed on the X-axis and the  $Y$  variable demand of bananas is placed on the Y-axis.

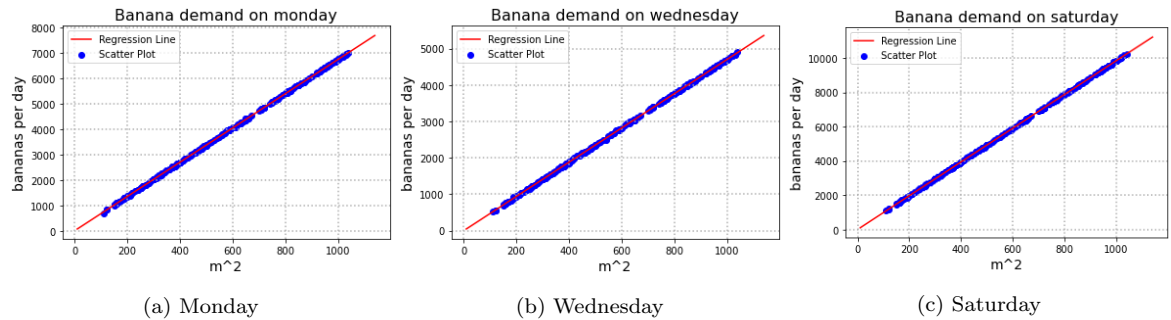


Figure 1: Demand for bananas on three different days



In Figure (1) it is clear to see that the data points fit the regression line almost perfectly for all the days, which means there is nearly no error term. Therefore, we can conclude there are no outliers. This can also be seen by measuring with  $R^2$ . When using OLS in Python, an  $R^2$  value of 1.000 is obtained for the data of each day. This result is also displayed in Table 1. This means that the size of a store explains all of the variation in the demand of bananas. The correlation between the two variables, demand of bananas and the size of a store, is definitely positive and linear. Specifically, this means that when the size of the store increases a fixed amount the demand of bananas also increases.

Since we have the size of the store as variable  $X$  and the demand of bananas as variable  $Y$ , we can predict the demand of bananas given a specific size of a store. The prediction of the demand of bananas are calculated by equation (3), in which we need the estimated intercept and slope. The estimated intercept and slope can be computed by using the method OLS, as described earlier. Equation (2) from methodology is used to compute these. We use a given sample size, which has  $n = 222$  observations of the variables  $X$  and  $Y$ . The computed estimates for every day are given in Table 1.

| Parameters          | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|---------------------|--------|---------|-----------|----------|--------|----------|
| $\widehat{\beta}_0$ | 6.146  | 1.668   | -2.010    | 2.459    | 1.767  | 4.709    |
| $\widehat{\beta}_1$ | 6.741  | 2.150   | 4.702     | 6.119    | 6.110  | 9.840    |
| $R^2$               | 1.000  | 1.000   | 1.000     | 1.000    | 1.000  | 1.000    |

Table 1:  $\widehat{\beta}_0$  and  $\widehat{\beta}_1$  for the different days.

When plugging in the obtained values of  $\widehat{\beta}_0$  and  $\widehat{\beta}_1$  in equation (2), the predicted value of the demand of bananas can be calculated for the days: Monday, Tuesday, Wednesday, Thursday, Friday and Saturday. Where the supermarket size  $X_i$  of every store, given in the data set, is also plugged in for calculating the predicted value of  $Y_i$ . When calculating this predicted value of the demand for bananas for every day and every store, the daily demand of bananas is obtained to which FL wants to provide its service. We have expressed the demand for bananas in roll pallets, with ten boxes of bananas per roll pallet. One box consists of fifty bananas. This is calculated for every supermarket chain separately. In Table 2, only the data for Monday is presented for the Junco supermarket chain. The abbreviations SS and DS represent supermarket size and demand supermarket, respectively. The file "Demand.xlsx" contains information about the daily demand for each supermarket chain.

To check for reliability in the results, we look at the assumptions of OLS, which are denoted in the methodology. The first assumption states that other factors contained in the error term are unrelated to the size of the store  $X_i$ . However, this assumption can be violated since there are factors that could potentially affect banana demand and may be correlated with store size. For example, marketing activities, such as discounts, can influence the demand. Larger stores may have more resources to invest in marketing. This may cast doubt on the results. The second assumption is satisfied, since it is clear to see that both the size of the store and the banana demand are independently and identically distributed. The third assumption is also satisfied since there are no outliers, as can be seen in Figure (1). Testing the extended least squares assumptions needs to be done in a specific order. First we test the assumption of homoskedasticity by using the Breusch-Pagan test. The residuals and the independent variable, which is the size of a store, are used as input in Python. The p-values for each different day have been obtained as output from the test and are presented in Table 3.

When using a significance level of 5 percent, we see that the p-value is higher than 0.05. This means there is no evidence to reject the null hypothesis and therefore no clear evidence for heteroskedasticity. Hence, we can see that assumption 4 is satisfied. Now, we can check assumption 5 by using the Jarque-Bera test. We calculated the test statistic value and the p-value in Python, and the results are presented in Table 4. In this table, a smaller test statistic value signifies that the sample data indicates characteristics that are more closely aligned with a normal distribution. We again compare the p-value of the Jarque-Bera test statistic with the significance level of 5 percent. All the p-values associated with the OLS test, shown in Table 4,

| City             | SS1<br>in $m^2$ | DS1<br>in $rp^*$ | SS2<br>in $m^2$ | DS2<br>in $rp^*$ | SS3<br>in $m^2$ | DS3<br>in $rp^*$ | SS4<br>in $m^2$ | DS4<br>in $rp^*$ | SS5<br>in $m^2$ | DS5<br>in $rp^*$ |
|------------------|-----------------|------------------|-----------------|------------------|-----------------|------------------|-----------------|------------------|-----------------|------------------|
| Alkmaar          | 652             | 8.80             | 997             | 13.45            | -               | -                | -               | -                | -               | -                |
| Amersfort        | 930             | 12.55            | -               | -                | -               | -                | -               | -                | -               | -                |
| Amsterdam        | 810             | 10.93            | 864             | 11.66            | 270             | 3.65             | 137             | 1.86             | 233             | 3.15             |
| Apeldoorn        | 868             | 11.71            | 990             | 13.36            | -               | -                | -               | -                | -               | -                |
| Arnhem           | 625             | 8.44             | 689             | 9.30             | 515             | 6.96             | -               | -                | -               | -                |
| Breda            | 842             | 11.36            | 580             | 7.83             | -               | -                | -               | -                | -               | -                |
| Delft            | 992             | 13.39            | 969             | 13.08            | 533             | 7.20             | -               | -                | -               | -                |
| Den Helder       | -               | -                | -               | -                | -               | -                | -               | -                | -               | -                |
| Dordrecht        | 611             | 8.25             | 795             | 10.73            | -               | -                | -               | -                | -               | -                |
| Ede              | 982             | 13.25            | 619             | 8.36             | -               | -                | -               | -                | -               | -                |
| Eindhoven        | 904             | 12.20            | 995             | 13.43            | 320             | 4.33             | 779             | 10.51            | -               | -                |
| Enkhuizen        | 908             | 12.25            | -               | -                | -               | -                | -               | -                | -               | -                |
| Gouda            | 850             | 11.47            | 801             | 10.81            | 494             | 6.67             | -               | -                | -               | -                |
| Haarlem          | 880             | 11.88            | 576             | 7.78             | 687             | 9.27             | 758             | 10.23            | -               | -                |
| Heerhugowaard    | -               | -                | -               | -                | -               | -                | -               | -                | -               | -                |
| Hoorn            | 924             | 12.47            | -               | -                | -               | -                | -               | -                | -               | -                |
| Leiden           | 874             | 11.80            | 523             | 7.06             | 350             | 4.73             | -               | -                | -               | -                |
| Medemblik        | -               | -                | -               | -                | -               | -                | -               | -                | -               | -                |
| Nieuwegein       | 758             | 10.23            | 570             | 7.70             | -               | -                | -               | -                | -               | -                |
| Nijmegen         | 986             | 13.31            | 610             | 8.24             | 225             | 3.05             | 132             | 1.79             | 215             | 2.91             |
| Oss              | 948             | 12.79            | 544             | 7.35             | 703             | 9.49             | -               | -                | -               | -                |
| Rotterdam        | 756             | 10.20            | 671             | 9.06             | 208             | 2.82             | 133             | 1.81             | 602             | 8.13             |
| Schagen          | -               | -                | -               | -                | -               | -                | -               | -                | -               | -                |
| 's-Hertogenbosch | 1002            | 13.52            | 619             | 8.36             | 127             | 1.72             | 261             | 3.53             | 658             | 8.88             |
| The Hague        | 805             | 10.87            | 604             | 8.16             | 484             | 6.54             | 754             | 10.18            | -               | -                |
| Tilburg          | 755             | 10.19            | 893             | 12.05            | 315             | 4.26             | 446             | 6.03             | 486             | 6.56             |
| Utrecht          | 885             | 11.94            | 693             | 9.36             | 434             | 5.86             | 720             | 9.72             | 782             | 10.56            |
| Wageningen       | 1007            | 13.59            | 908             | 12.25            | -               | -                | -               | -                | -               | -                |
| Zaandam          | -               | -                | -               | -                | -               | -                | -               | -                | -               | -                |

Table 2: Monday's Rounded Demand for Junco. (\*  $rp$  = roll pallets)

| Parameters | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|------------|--------|---------|-----------|----------|--------|----------|
| p-value    | 0.232  | 0.634   | 0.550     | 0.467    | 0.386  | 0.998    |

Table 3: Results of the Breusch-Pagan test using OLS.

are greater than 0.05. Hence, there is no evidence to reject the null hypothesis, which means there is no evidence to state that the skewness and kurtosis are different from a normal distribution. We can say that assumption 5 is satisfied. So, due to assumption 1 only we can not guarantee an unbiased, consistent, asymptotically normal and efficient OLS estimator.

| Parameters           | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|----------------------|--------|---------|-----------|----------|--------|----------|
| test-statistic value | 0.145  | 4.044   | 0.068     | 2.387    | 4.067  | 1.292    |
| p-value              | 0.930  | 0.132   | 0.966     | 0.303    | 0.131  | 0.524    |

Table 4: Results of the Jarque-Bera test using OLS.

We already saw that the regression line approximates the real data points perfectly with OLS, but we still wanted to compare this method with WLS. Measuring with  $R^2$  for WLS, gives the same value of 1.000 for each day in the week. The size of a store explains all of the variation in

the demand of bananas. We apply the same tests as OLS on WLS. The results obtained from the Breusch-Pagan test are provided in Table 5. The p-values obtained from the WLS are very similar to those from the OLS, with slight differences. All p-values are above the significance level of 5 percent, indicating that the null hypothesis cannot be rejected. Applying the Jarque-Bera test in Python yields the following results, as presented in Table 6. These p-values also slightly differ compared to the values obtained when OLS. In this case, once again, all the p-values are above the 5 percent significance level. Therefore, the null hypothesis cannot be rejected. So, the outcomes of both methods are nearly the same. However, it is known that a smaller p-value signifies stronger evidence. Thus, it can be stated that when utilizing the WLS method, the evidence is stronger for certain days, but not necessarily for all days.

| Parameters | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|------------|--------|---------|-----------|----------|--------|----------|
| p-value    | 0.232  | 0.637   | 0.550     | 0.467    | 0.384  | 0.999    |

Table 5: Results of the Breusch-Pagan test using WLS.

| Parameters           | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|----------------------|--------|---------|-----------|----------|--------|----------|
| test-statistic value | 0.368  | 3.575   | 0.006     | 1.784    | 3.827  | 1.274    |
| p-value              | 0.832  | 0.167   | 0.997     | 0.410    | 0.148  | 0.529    |

Table 6: Results of the Jarque-Bera test using WLS.

## Total costs

### Non-cooperative

To go further in the analysis for Fruit Logistics and determine the optimal sequence for delivering their offerings to supermarkets, it is essential to explore the data in greater detail. The demand for each supermarket within a given supermarket chain is currently being assessed in terms of roll pallets. Now the objective is to calculate the total costs that each supermarket has from individually storing and transporting the bananas. This is non-cooperative. These costs are represented as  $C_0(\{AH\})$  for the Albert Hijné,  $C_0(\{J\})$  for the Junco, and  $C_0(\{K\})$  for the Koop.

The data at our disposal includes various supermarkets located in different cities of the different supermarket chains, the travel time and the distance between all the cities, and the capacity of the trucks of the corresponding supermarket chain. The costs of various elements are also known: the fixed costs of the trucks, the variable costs of driving per hour, the costs per km, and the storage costs that come with the storage of the bananas.

This scenario can be seen as a VRP. The vehicle routing problem represented by this scenario is a CVRP, given that the trucks available to this problem have a specific limited capacity. It is not a VRP with time windows, as we have not been provided with time slots for our customers (the supermarkets). Additionally, it is not a multi-depot VRP, as each supermarket chain only has one depot.

If the demand exceeds the combined capacity of all the trucks at our disposal, the trucks need to make multiple trips in a day. This is because the truck routes do not fill an entire working day for a driver due to capacity constraints. In such a case, it entails dealing with a multi-trip VRP. This is also in our scenario the case.

To calculate the individual costs for each supermarket chain, it is crucial to compute both the total distance and the total time involved. Both factors contribute to costs as the total distance incurs expenses due to driving costs, such as gasoline, of the truck, and the time is a significant factor since drivers need to be compensated for their work. Our aim is to minimize the total distance, as doing will also minimize the driving and unloading hours in our case. This strategy is viable because we currently do not face any time constraints. This differs from a VRP with time windows, where specific time restrictions on deliveries must be taken into consideration.

To calculate these costs, the optimal routes to deliver the bananas to all supermarkets needs to be determined. In our case, our VRP is structured as follows: the vehicles represent our trucks, and the customers are all the supermarkets to which we deliver.

The VRP we have is a CVRP, this can be formulated as a linear integer programming model. It can be formulated following the model of Kim (2021). To construct this model with the given constraints, we first initialize certain variables.

1.  $G = (V, E)$  : A graph showing the location and route of the customer
2.  $V = \{0, 1, \dots, n\}$ : A collection of nodes. The node that represents the depot is 0, and the nodes that represent each customer are 1 to  $n$ .
3.  $E$  : Set of Edges connecting each node  $e_{ij} = (i, j)$
4.  $K = \{1, 2, \dots, |K|\}$  : Set of vehicles
5.  $d_i \geq 0$  : demand of customer  $i$
6.  $Q \geq 0$  : Maximum vehicle capacity (all vehicles have the same maximum capacity  $Q$ )
7.  $c_{ij}$  : Travel costs between customer  $i$  and  $j$  (distance between  $i$  and  $j$ )

We also initialize a decision variable. This variable indicates whether the vehicle  $k$  has taken the route  $e_{ij}$  :

$$x_{ij}^k = \begin{cases} 1, & \text{taken.} \\ 0, & \text{not taken.} \end{cases} \quad (11)$$

The model is as follows:

1. 
$$\min \sum_{k \in K} \sum_{(i,j) \in E} c_{ij} x_{ij}^k$$
2. 
$$\sum_{k \in K} \sum_{i \in V, i \neq j} x_{ij}^k = 1 \quad \forall j \in V \setminus \{0\}$$
3. 
$$\sum_{j \in V \setminus \{0\}} x_{0j}^k = 1 \quad \forall k \in K$$
4. 
$$\sum_{i \in V, i \neq j} x_{ij}^k - \sum_{i \in V} x_{ij}^k = 0 \quad \forall j \in V, \forall k \in K$$
5. 
$$\sum_{i \in V} \sum_{j \in V \setminus \{0\}, i \neq j} q_j x_{ij}^k \leq Q \quad \forall k \in K$$
6. 
$$\sum_{k \in K} \sum_{(i,j) \in S, i \neq j} q_j x_{ij}^k \leq |S| - 1 \quad S \subseteq V \setminus \{0\}$$
7. 
$$x_{ij}^k \in \{0, 1\} \quad \forall k \in K, \forall (i, j) \in E$$

The constraints are formulated for the decision variable and the objective function of CVRP (Kim, 2021). All the constraints are explained as follows:

1. The first thing (1) that can be seen in the model is the objective function, where we want to minimize the travel costs of all vehicles.

2. The constraint at (2) implies that each customer location can only be visited once by a single vehicle.
3. The constraint at (3) means that the routes need to have the depot as their starting location.
4. The constraint at (4) states that the quantity of vehicles entering and exiting a customer's locations needs to be the same.
5. The constraint at (5) specifies that the delivery capacity of each vehicle must not surpass the vehicle's maximum capacity.
6. The constraint at (6) ensures the elimination of subtours within the solution, preventing the formation of disconnected or incomplete routes in the overall solution.
7. This last constraint (7) indicates that the variable is a binary variable, taking the values of either 0 or 1 to represent the two possible states or choices.

This problem cannot be solved manually. Therefore, we require an algorithm capable of addressing it. To tackle our case, we employ a high-performance implementation of the HGS, explained in the methodology. In our case, we apply the algorithm with the stopping criteria set as a specified number of iterations.

Before we proceed to solve this problem, a few important statements and assumptions need to be clarified before we can do the calculations.

The first statement is that for a feasible solution to the CVRP the set of routes needs to all begin and end at the depot, which was already stated in the second constraint 3. This ensures that each client is visited exactly once, and no route exceeds the vehicle capacity ( $pyV, nd$ ).

An assumption that we make is that we do not consider loading time for the trucks. We assume that the trucks are already loaded. Without this assumption, it would be hard to calculate the costs, as loading time would consume time and incur expenses, but we do not have the corresponding pricing and timing information.

The next assumption we make is that a truck driver requires 11 hours of rest daily, resulting in a working day of 13 hours. Within these working hours, the driver is allowed to drive the vehicle up to 9 hours per day (Waterstaat, 2023). We will use this information later in the calculations to optimize the entire process.

Our last assumption that we make is that we do not have to deliver whole pallets to a store. This means that when delivering to a store, we are not obligated to unload entire roll pallets; instead, we can deliver entire pallets along with a few boxes from other pallets. This approach allows for more effective optimization at the moment, as it eliminates the need to transport partially filled roll pallets, enabling us to maximize the loading capacity of trucks. As a result, we use the demand rounded to two decimal places, as demonstrated in Table 2 for Junco on Monday.

Now, we can proceed with the actual calculations of the individually storing and transporting costs of the bananas of the different supermarket chains. The model we have created in Python with the HGS will optimize routes, taking into account both the fixed costs of the vehicles and the distances of the routes. The model that we use is designed for only one route per vehicle, meaning each vehicle can only undertake a single route. In our scenario, we are dealing with a multi-trip VRP, which requires us to do multiple routes per vehicle. This is essential to accommodate the volume of demand and capacity effectively, as a single route per vehicle would not be sufficient. As a result, we initially estimated the number of vehicles that we need to use based on the working time and driving time, using the working hours from our assumption. To ultimately determine the number of vehicles, we start by selecting the maximum of the number of vehicles we needed to achieve the 13 hours of working time and the number of vehicles required to meet the 9 hours of travel time, ensuring not to exceed the higher of the two. As we aim to avoid imposing impossibly long working days on the driver. Once we obtain this maximum, we then consider the minimum between this number of vehicles and the total number of vehicles available. Subsequently, we allocated the routes among this number of vehicles. In this manner, we can

attempt to optimize the number of vehicles. For instance, if a supermarket has more vehicles at its disposal than necessary to fill entire workdays, it could result in unnecessary costs.

After the execution of this program for a VRP several Key Performance Indicators (KPIs) are generated. The KPIs include the number of planned routes, the number of planned stores, the total distance, and the total driving time. They are all displayed in "kips.xlsx".

To illustrate a few KPIs and showcase what a route looks like, we select a specific supermarket chain and day. In this example, we focus on the Junco supermarket on Monday. A selection of essential KPIs, from "kips.xlsx", for Junco on Monday is presented in Table 7.

|                                                   |        |
|---------------------------------------------------|--------|
| <b>Company</b>                                    | Junco  |
| <b>Day</b>                                        | Monday |
| <b>Feasible</b>                                   | TRUE   |
| <b>Distance (in km)</b>                           | 4158,6 |
| <b>Number of routes</b>                           | 23     |
| <b>Sum of demand (in roll pallets)</b>            | 659,17 |
| <b>Travel duration (in min)</b>                   | 3515   |
| <b>Service duration (unloading time) (in min)</b> | 923    |
| <b>Number of vehicles</b>                         | 4      |

Table 7: KPIs of Junco on Monday

For this supermarket on this particular day, it can be observed that the working hours are quite extensive. The total working time is calculate by adding the travel duration to the service duration and then dividing the result by the total number of vehicles.

$$Total\ working\ time\ (in\ min) = \frac{Travel\ duration + Service\ duration}{Number\ of\ vehicles}$$

$$Total\ working\ time\ (in\ min) = \frac{3515 + 923}{4} = 1109.5$$

In this scenario, the driver is required to work for 1109.5 minutes, equivalent to 18.49 hours. This constitutes an exceptionally long working day, which may not be feasible, according to Waterstaat (2023) and the thus the assumption we have made. However, in this scenario, we are constrained as we cannot increase the number of vehicles since Junco only has 4 vehicles.

An example of such a route generated by the algorithm is visible in the Figure 2. Here, you can observe the route for Junco on Monday. The routes for all days for each scenario have been added to the appendix. The corresponding sequence of cities for transportation is presented in the "routesheets.xlsx" file.

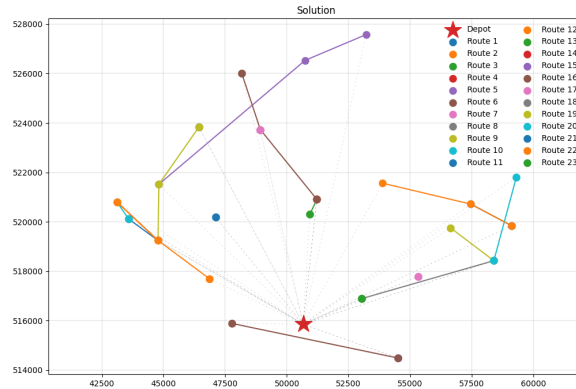


Figure 2: Route for Junco on Monday

There are several elements that we need to calculate the total costs, namely the distance costs, the time costs, the storage costs, and the vehicle costs. We have been provided with the information of Table 8. We will use this information in our calculations.

| Transporter          | Cost per km<br>(in €) | Cost per hour<br>(in €) | Storage cost*<br>(in €) | Fixed vehicle<br>cost (in €) |
|----------------------|-----------------------|-------------------------|-------------------------|------------------------------|
| Junco (J)            | 1,1                   | 10,4                    | 5,9                     | 113                          |
| Albert Hijne (AH)    | 0,9                   | 11                      | 6                       | 111                          |
| Koop (K)             | 0,9                   | 10,6                    | 5,8                     | 115                          |
| Fruit Logistics (FL) | 1                     | 10,2                    | 5,4                     | 120                          |

Table 8: The provided costs per transporter. \*(per roll per day)

The formula of the total costs is as follows:

$$Total\ costs = \sum_{i=0}^5 (Distance\ costs_i + Time\ costs_i + Storage\ costs_i + Vehicle\ costs_i) \quad (12)$$

with  $i$  being the number of the day, starting with 0 for monday and ending with 5 for saturday.

First, we will calculate the distance costs using the following formula:

$$Distance\ costs = Total\ distance * Cost\ per\ km \quad (13)$$

Where the total distance is a KPI of our program, and the costs per kilometer are provided in Table 8.

The next set of costs we will calculate are the costs for time:

$$Time\ costs = (Total\ driving\ time + Total\ unloading\ time\ per\ roll) * Cost\ per\ hour \quad (14)$$

Where the driving time is a KPI, the cost per hour is specified in Table 8. The unloading time per roll pallet is determined by the following formula, with a given unloading time of 1.4 minutes for each roll pallet.

$$Total\ unloading\ time\ per\ roll = Total\ demand * 1.4 \quad (15)$$

Next, we also want to calculate the vehicle costs. For this, we first determined the total working time for the drivers, considering both driving and unloading time of the roll pallets. Additionally, we assessed the total driving time for the drivers. The total working time is divided by 13, as a driver is allowed to work up to 13 hours. Similarly, the total driving time is divided by 9, reflecting the maximum driving allowance per day, as mentioned in a assumption before. We take the maximum of these two values to ensure that we use enough vehicles for the planned routes, avoiding overworking or excessive driving for any individual driver. This estimation helps us determine the required number of vehicles and, consequently, estimate the costs for the vehicles. From this, the following formula for the costs of the vehicles can be derived:

$$Vehicle\ costs = Number\ of\ vehicles * Fixed\ vehicle\ cost \quad (16)$$

The last costs we want to calculate are the storage costs:

$$Storage\ cost = Total\ demand * Storage\ cost\ per\ day * Storage\ duration\ in\ days \quad (17)$$

We make the following assumption: we store the bananas for only one day. We do this because bananas cannot be stored for an extended period as they may rot or turn brown. Therefore our goals is to minimize the storage duration for bananas. We incorporate this assumption into our further calculations. As a result, the Storage duration in days is set to one day.

So, to calculate the individual costs per supermarket, we plug in these formulas with all the data we have received and calculated. Once we have completed this process, the following results will emerge, in Table 9.

| Scenario Number | Scenario      | Transporter  | Sum of distance (in km) | Sum of total cost (in €) |
|-----------------|---------------|--------------|-------------------------|--------------------------|
| 0               | $C_0(\{J\})$  | Junco        | 22884                   | 52686                    |
| 1               | $C_0(\{AH\})$ | Albert Hijne | 28569                   | 62456                    |
| 2               | $C_0(\{K\})$  | Koop         | 14784                   | 38770                    |

Table 9: The rounded total costs for each supermarket chain individually.

### Cooperative

After calculating the total costs for individual stores, Junco, Albert Hijne, and Koop, we will also perform the same analysis considering that we supply everything from Fruit Logistics. This collaborative approach provides additional information in our further analysis to determine the best approach for Fruit Logistics.

We will address various scenarios. Initially, we will examine the costs associated with Fruit Logistics supplying one supermarket. So, Fruit Logistics supplying Junco (J) separately, Albert Hijne (AH) separately, and the Koop (K) separately. After completing these various scenarios, there are additional ones to consider. Specifically, Fruit Logistics supplying multiple supermarkets simultaneously. The scenarios include (J, AH), (J, K), (AH, K), and finally, we also have the scenario where Fruit Logistics supplies all three supermarkets (J, AH, K). So, we aim to calculate the costs for the following scenarios:  $C(\{J\})$ ,  $C(\{AH\})$ ,  $C(\{K\})$ ,  $C(\{J, AH\})$ ,  $C(\{J, K\})$ ,  $C(\{AH, K\})$ , and  $C(\{J, AH, K\})$ .

For these seven scenarios, we can utilize the same HGS algorithm to determine the optimal route and compute the cost, as we did when determining the individual costs,  $C_0(\{J\})$ ,  $C_0(\{AH\})$ ,  $C_0(\{K\})$ . The model utilizes the locations of all the supermarkets in the given scenario and the costs provided in Table 8 for Fruit Logistics to perform the calculations.

An example of a route planned by Fruit Logistics for Junco and Albert Hijne on Monday is illustrated in Figure 3.

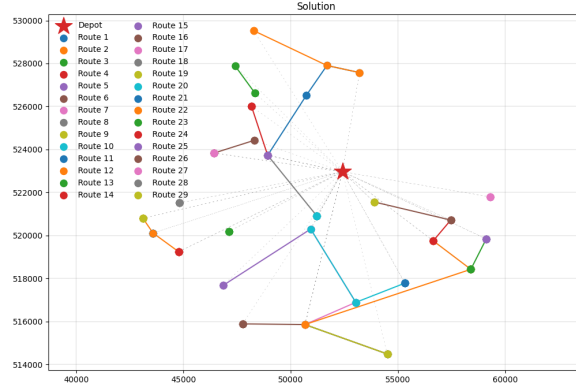


Figure 3: Route for J and AH on Monday by FL

The following results about the distance and costs are derived from the program we utilized. In Table 10, we merged the previous scenarios with the new ones, allowing for a comparative analysis.



| Scenario Number | Scenario          | Transporter     | Sum of distance (in km) | Sum of total cost (in €) |
|-----------------|-------------------|-----------------|-------------------------|--------------------------|
| 0               | $C_0(\{J\})$      | Junco           | 22884                   | 52686                    |
| 1               | $C_0(\{AH\})$     | Albert Hijne    | 28569                   | 62457                    |
| 2               | $C_0(\{K\})$      | Koop            | 14784                   | 38770                    |
| 3               | $C(\{J\})$        | Fruit Logistics | 13654                   | 38893                    |
| 4               | $C(\{AH\})$       | Fruit Logistics | 17600                   | 50928                    |
| 5               | $C(\{K\})$        | Fruit Logistics | 11273                   | 34736                    |
| 6               | $C(\{J, AH\})$    | Fruit Logistics | 28385                   | 85380                    |
| 7               | $C(\{J, K\})$     | Fruit Logistics | 22986                   | 70521                    |
| 8               | $C(\{AH, K\})$    | Fruit Logistics | 26325                   | 81644                    |
| 9               | $C(\{J, AH, K\})$ | Fruit Logistics | 37461                   | 116387                   |

Table 10: The rounded total costs for each scenario.

### Cost savings

After calculating all the costs,  $C(S)$ , where  $S \subseteq \{J, AH, K\}$ , for the different scenarios, we will proceed to calculate the cost savings. This will assist us in providing advice to Fruit Logistics regarding the optimal order in which to approach the stores. The formula for the cost savings is given as follows:

$$v(S) = P_{i \in S} C_0(\{i\}) - C(S), \quad S \subseteq \{J, AH, K\} \quad (18)$$

Here,  $C_0(\{i\})$  represents the cost of supplying store  $i$  individually. All the cost savings for each scenario are provided in Table 11.

| Scenario Number | Scenario          | Sum of total cost (in €) | Cost savings (in €) |
|-----------------|-------------------|--------------------------|---------------------|
| 0               | $C_0(\{J\})$      | 52686                    | 0                   |
| 1               | $C_0(\{AH\})$     | 62457                    | 0                   |
| 2               | $C_0(\{K\})$      | 38770                    | 0                   |
| 3               | $C(\{J\})$        | 38893                    | 13792               |
| 4               | $C(\{AH\})$       | 50928                    | 11528               |
| 5               | $C(\{K\})$        | 34736                    | 4034                |
| 6               | $C(\{J, AH\})$    | 85380                    | 29762               |
| 7               | $C(\{J, K\})$     | 70521                    | 20934               |
| 8               | $C(\{AH, K\})$    | 81644                    | 19583               |
| 9               | $C(\{J, AH, K\})$ | 116387                   | 37526               |

Table 11: The rounded total costs and cost savings for each scenario.

Note that in Table 9, 10, and 11, the costs are rounded to the nearest whole number. However, to calculate the cost savings, we utilized the non-rounded numbers. Therefore, if you perform the calculations of the cost savings using the numbers of the costs from the table, the result may differ slightly from what is stated in the table under cost savings.

### Maximum Cost Saving

After calculating the cost savings from all the scenarios, we propose two games to advise FL on how to allocate the cost savings among the supermarket chain coalition. Before we delve into the games, it is important to consider a few assumptions. Firstly, each supermarket has the option to either accept or reject FL's offer twice. Secondly, we introduce another transportation company, VL, which differs from FL in its approach. VL does not approach supermarkets sequentially; instead, it presents all information about the share of cost savings at once, distributing these

savings based on the Shapley Value. VL makes its offer to the supermarkets following FL's initial proposal. Thirdly, during the second round of decision-making, a supermarket will accept FL's offer if it is higher than VL's; if it's lower, they will reject FL and opt for VL. However, the supermarkets are unaware of each other's choices regarding FL's offer. Finally, FL is bound to its initial offer and cannot modify it once presented to the supermarkets.

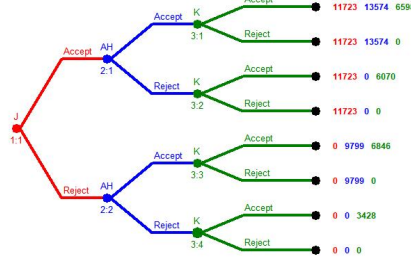


Figure 4: Illustration of the Sequential Game with  $\pi = (J, AH, K)$

We have developed a sequential game involving three players  $J, AH, K$  for each permutation  $\pi$ , where the players can choose between the strategies 'Accept' and 'Reject'. The decision nodes are all singletons since we assume that there is perfect information. We suppose that if a chain accepts the service of FL, the other chains will know due to trucks driving around or from advertisements.

Before diving into the game, it is notable to mention whether the assumptions, which are mentioned in the methodology part, are satisfied for extensive game. First, the assumption of consistent predictions seems to be satisfied as players make decisions based on FL's offer, which is predictable and consistent. With perfect information about the game's sequence, players can coordinate towards a common equilibrium, satisfying the second assumption. The assumption of self-fulfilling predictions is also met as the player's decision to accept FL's offer due to positive cost savings seems to be a self-fulfilling outcome. The fact that the players are acting rationally in response to the offer from FL and there is no something interrupt this implementation of strategies also satisfies the fourth assumption. Lastly, we assume players to understand the game and make decisions rationally, meeting the last assumption.

Given that there are six such games with identical structures, we will only discuss one of these games here. The game, depicted in Figure 4, follows the order  $J - AH - K$ . In this scenario, the value each player (supermarket) receives is determined by FL's initial offer. It's important to note that this offer is calculated as 85% of the cost savings generated by the coalitions. By employing backward induction equilibrium, to determine the game's equilibrium, we find that the most favorable outcome for all players is to accept FL's offer. This decision is logical since the cost savings are positive, making the offer attractive to all supermarkets.

Then, to figure out which supermarket would like to accept or reject, a cooperative game is introduced, as explained in the methodology. This game again consists of a finite of players  $N = 3$  who cooperate to share the cost savings. A characteristic function  $v(S)$ , for cost savings, which assigns a value to each coalition, is defined in 19. We have taken here the cost savings from the previously mentioned Table 11 and multiplied them by 0.85% to assess FL's potential actions.

$$v(S) = \begin{cases} 11724, & \text{if } S = \{J\} \\ 9799, & \text{if } S = \{AH\} \\ 3429, & \text{if } S = \{K\} \\ 25298, & \text{if } S = \{J, AH\} \\ 17794, & \text{if } S = \{J, K\} \\ 16645, & \text{if } S = \{AH, K\} \\ 31897, & \text{if } S = \{J, AH, K\} \end{cases} \quad S \subseteq \{J, AH, K\} \quad (19)$$

|               | J     | AH    | K    |
|---------------|-------|-------|------|
| Shapley Value | 13969 | 12432 | 5495 |

Table 12: Shapley Value

| $\pi$    | $\sigma(J, AH, K)$   | supermarket that accepts | extra | cost savings of FL |
|----------|----------------------|--------------------------|-------|--------------------|
| (J,AH,K) | (11724, 13574, 6598) | AH,K                     | 3527  | -589               |
| (J,K,AH) | (11724, 14103, 6070) | AH,K                     | 3527  | -589               |
| (AH,J,K) | (15499, 9799, 6599)  | J,K                      | 4303  | -1162              |
| (AH,K,J) | (15252, 9799, 6846)  | J,K                      | 4303  | -1162              |
| (K,J,AH) | (14365, 14103, 3429) | J,AH                     | 3170  | 1294               |
| (K,AH,J) | (15252, 13216, 3429) | J,AH                     | 3170  | 1294               |

Table 13: The Outcome of the Second Decision

To determine which supermarkets are likely to accept or reject the offer, we utilize the Shapley Value table, as referenced in Table 12. This table is derived using the formula mentioned in 10. VL, the competing company, employs these values to propose offers to the three supermarkets. Each supermarket then compares these values with FL's initial offers to decide whether to continue accepting or to reject. Table 13 outlines the outcomes of their second decision.

The opening offer received by each supermarket, denoted as  $\delta(J, AH, K)$ , is presented according to their respective order. For instance, in the order (J, AH, K), supermarkets J, AH, and K receive offers of  $v(J)$ ,  $v(J, AH) - v(J)$ , and  $v(J, AH, K) - v(J, H)$  respectively. It's important to note that FL's offer to each supermarket varies based on the order  $\pi$ , which implies that the order could influence which supermarket group accepts the offer in the second round. The third column of the table clarifies which supermarkets adhere to their offers. The final column shows the additional cost that FL must incur from the cost savings, as some supermarkets may opt to reject, and FL is unable to alter their initial offer. Thus, FL's goal is to minimize this additional expense.

Consequently, FL should strategically negotiate with the supermarkets in the sequences of (K, J, AH) and (K, AH, J), with offers of (14365, 14102, 3428) and (15251, 13216, 3428) respectively. By adopting this strategy, FL stands to achieve a cost saving of 1294. In Scenario 6, the optimal outcome occurs with the formation of coalition  $S = J, AH$ .

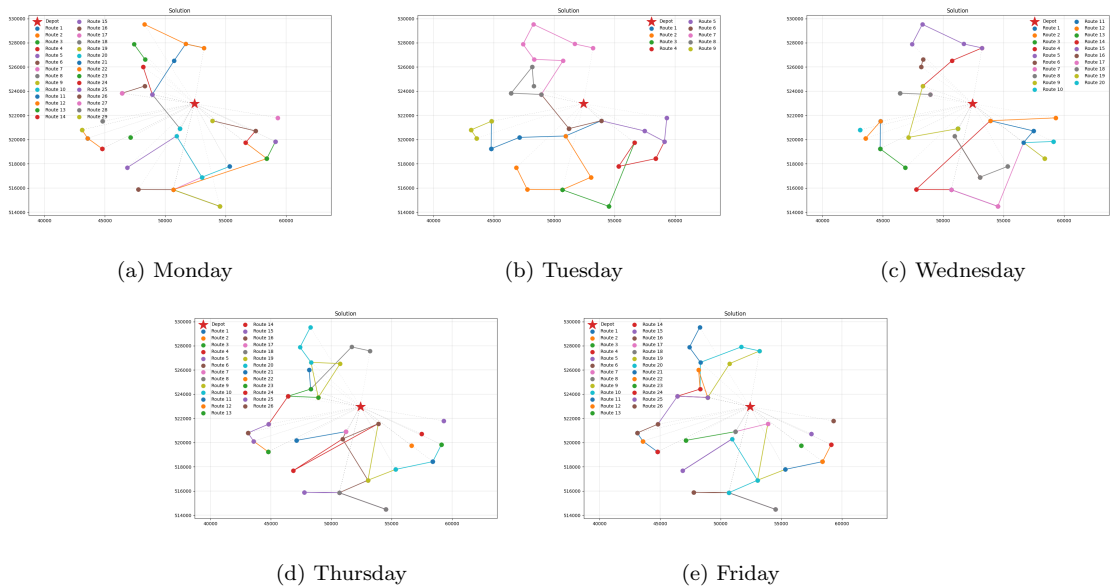


Figure 5: Routes for scenario 6 where  $S = J, AH$ .

Figure 5 illustrates the five working days planning for the supermarket group of J,AH.

## Critical Review

### Discussion

Several key discussion points need to be addressed. We employed the HGS algorithm in Python, which terminates when a specific criterion is reached. Initially, we set the criterion to a time limit and observed that an optimal route solutions were achieved after a certain number of iterations. Subsequently, we adjusted the criterion to that iteration count, allowing us to consistently replicate the same routes with each run. It is worth noting that there is a possibility of obtaining a better solution if the program is allowed to run for a longer duration, allowing for more time and a higher number of iterations.

The second point for discussion is our handling of demand, which was rounded to two decimal places in our calculations. There is a potential of achieving an improved solution by rounding the demand not just to two decimal places but rather up to boxes of bananas. Assuming that we can deliver individual boxes from the roll pallets instead of entire roll pallets. This approach ensures that each supermarket receives a sufficient quantity of bananas, possible with a slight surplus.

The next point is that the HGS model we used is designed for only one route per vehicle, as is mentioned before, however we have a multi-trip. As mentioned earlier before the calculations happened, we estimated the number of vehicles and selected the minimum. This impacts both the number of vehicles we utilize and, consequently, the costs associated with them. While the difference may not be significant, if the model had been designed for multiple trips, it could potentially yield a better solution, by optimizing the number of vehicles. This is because it would have the capability to optimize the overall fleet more effectively.

Finally, while we introduce another company, VL, to determine the optimal order, it is important to note that, in real-world multiple companies would likely be involved. Furthermore, for FL to maximize cost savings, a sequential approach to each supermarket would be not the most efficient strategy. Instead, providing comprehensive information about all potential cost savings from the outset would be more beneficial. This approach ensures that all supermarkets are likely to accept, given that the cost savings are positive and advantageous for each.

### Conclusion

Fruit Logistics aims to engage with three supermarket chains Albert Hijne (AH), Junco (J), and Koop (K), in the Netherlands for transportation, with a specific focus on transporting bananas. The objective of FL is for the supermarket chains to entrust their transportation to FL. The company intend to achieve this by offering the supermarket chains a share of the cost savings. FL aims to approach these supermarkets in an optimal order, ensuring that the earnings for FL are maximized.

To establish the order for approaching supermarkets, we undertook several calculations. Our process started with determining the demand, followed by seeking an optimal solution for the VRP by defining routes with a HGS algorithm. With the results of the algorithm, we computed the total costs for each scenario. Subsequently, we calculated the cost savings.

The most significant findings from this report are related to the cost savings. We found that when FL handles the transportation of bananas, it turns out to be more cost-effective than if the supermarket chains handle the transportation themselves.

To achieve maximum cost savings, there are two optimal orders for approaching these three supermarket chains: the first is Koop first, followed by Junco and Albert Hijne, and the second is Koop first, then Albert Hijne, and finally Junco. In these orders, Junco and Albert Hijne agree to the services, and the five working days plan is established for this group of supermarkets.

## References

- (n.d). A brief introduction to vrp.
- Baker, B. M. and M. Ayechew (2003). A genetic algorithm for the vehicle routing problem. *Computers Operations Research* 30(5), 787–800.
- Bertsimas, D. and J. N. Tsitsiklis (1997). *Introduction to Linear optimization*. Athena Scientific.
- Houba, H. (2023). *Lecture Notes Mathematical Economics 1*.
- Kim, K. (2021, Aug). Capacitated vehicle routing problem (cvrp) with python+pulp and google maps api.
- Konstantakopoulos, G. D., S. P. Gayialis, and E. P. Kechagias (2020). Vehicle routing problem and related algorithms for logistics distribution: a literature review and classification. 22, 2033–2062.
- Michael Maschler, Eilon Soan, S. Z. (2020). Game theory.
- Rensink, R. A. and G. Baldrige (2010). The perception of correlation in scatterplots. *Computer Graphics Forum* 29(3), 1203–1210.
- Savani, R. and T. L. Turocy (2024). Gambit: The package for computation in game theory.
- Stock, J. H. and M. W. Watson (2020). *Introduction to econometrics*. Pearson.
- Toth, P. and D. Vigo (2002). *The Vehicle Routing Problem*.
- Vidal, T. (2022). Hybrid genetic search for the cvrp: Open-source implementation and swap\* neighborhood. *Computers amp; Operations Research* 140, 105643.
- Waterstaat, M. v. I. e. (2023, Nov). Wat is er wettelijk geregeld voor de rijtijden en rusttijden bij wegvervoer?
- Yu, Y., X. Wang, R. Y. Zhong, and G. Q. Huang (2016). E-commerce logistics in supply chain management: Practice perspective. *Procedia CIRP* 52, 179–185. The Sixth International Conference on Changeable, Agile, Reconfigurable and Virtual Production (CARV2016).

# Appendix

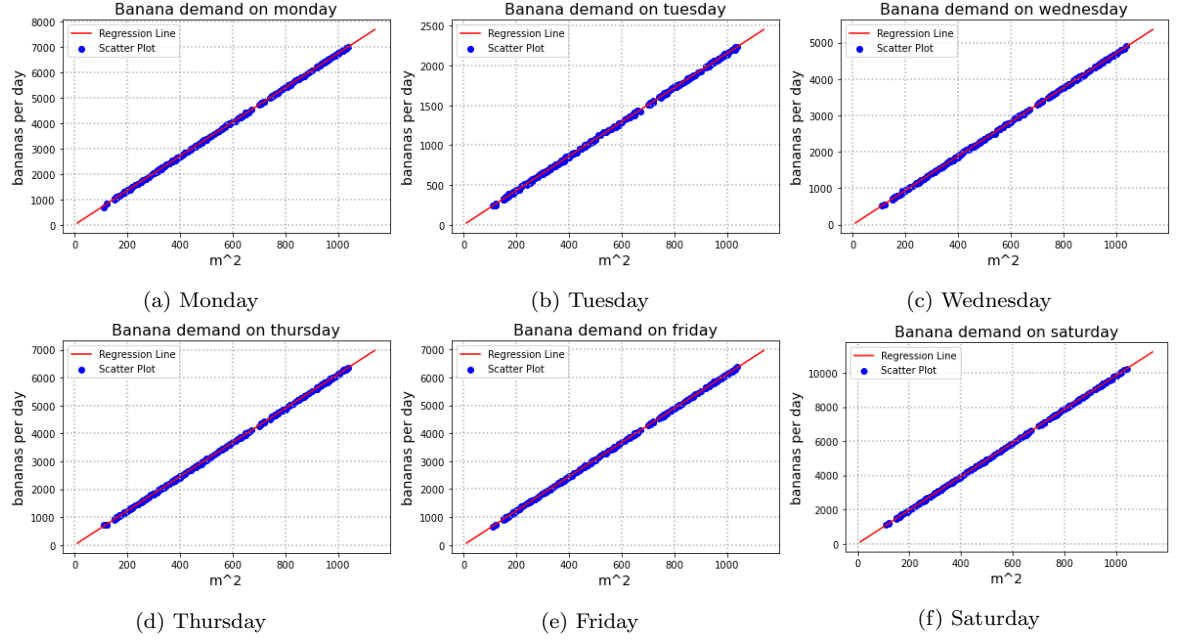


Figure 6: Demand for bananas on each day of the week

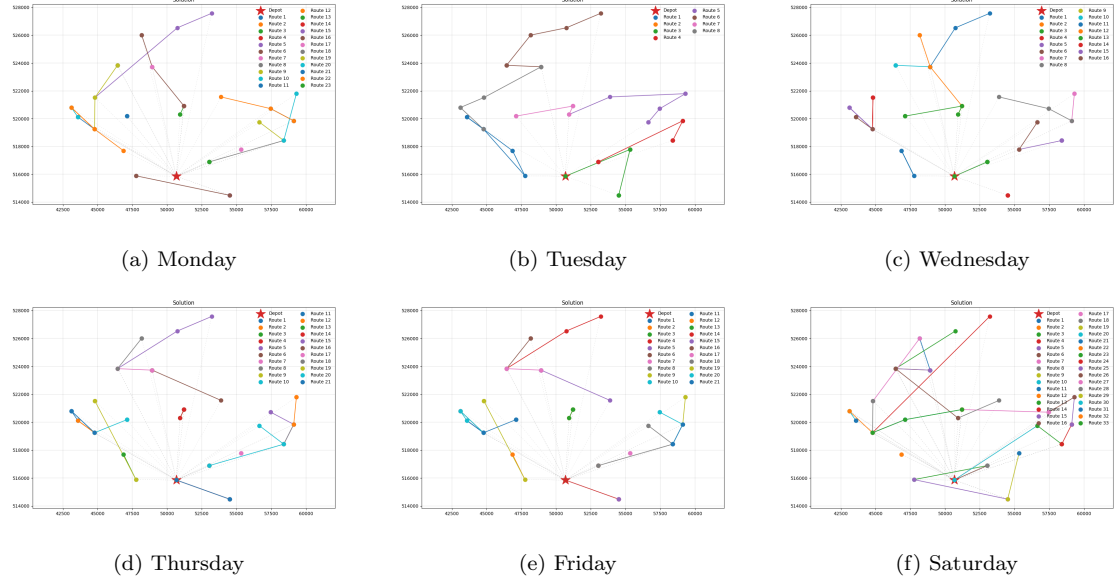
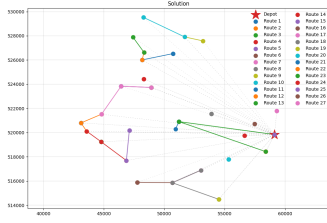
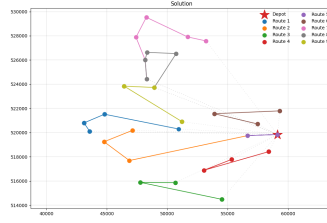


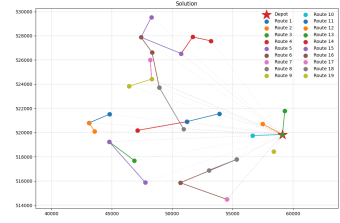
Figure 7: Routes for scenario 0 (J individually).



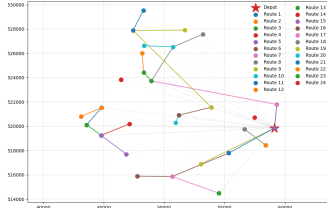
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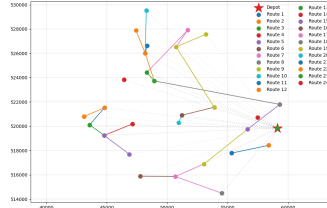
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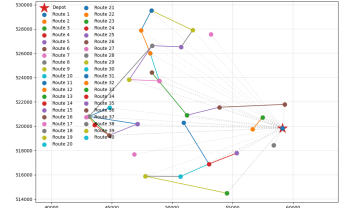
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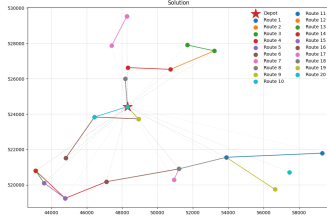


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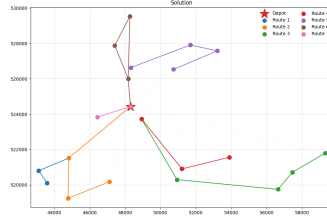


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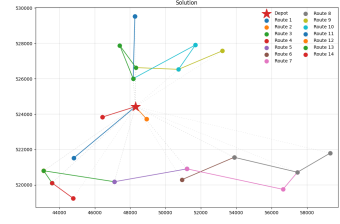
Figure 8: Routes for scenario 1 (AH individually).



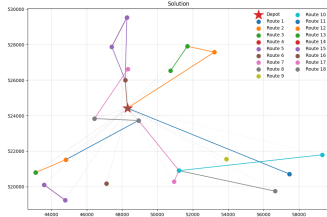
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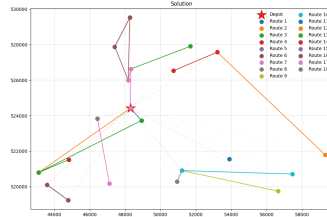
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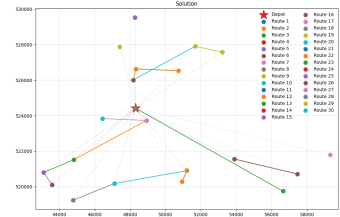
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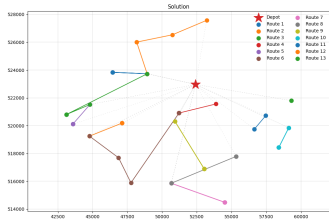


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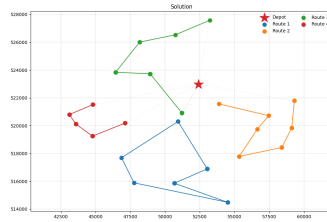


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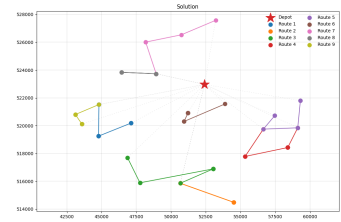
Figure 9: Routes for scenario 2 (K individually).



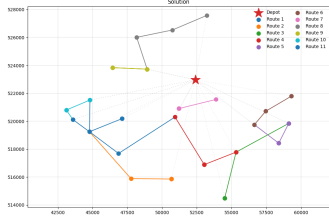
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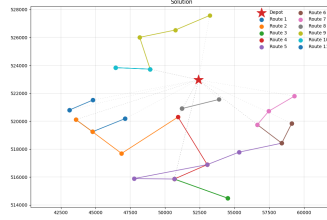
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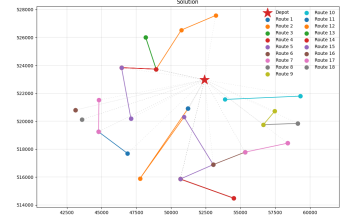
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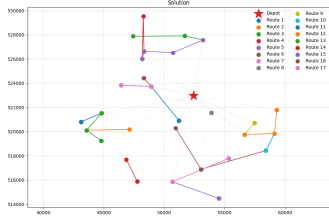


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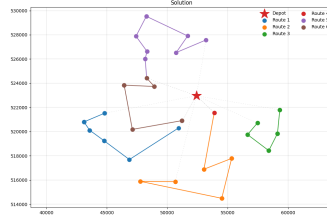


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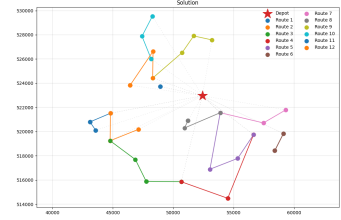
Figure 10: Routes for scenario 3 (J by FL).



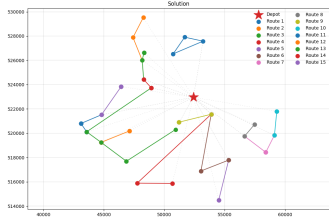
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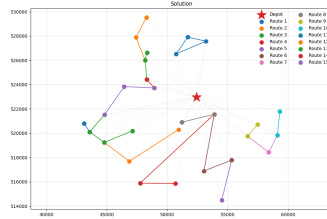
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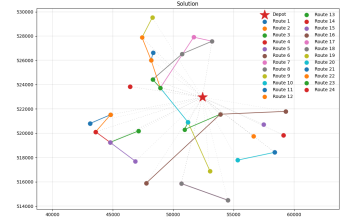
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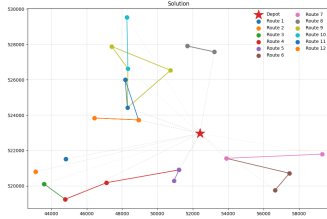
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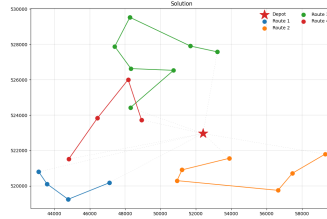
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Figure 11: Routes for scenario 4 (AH by FL).

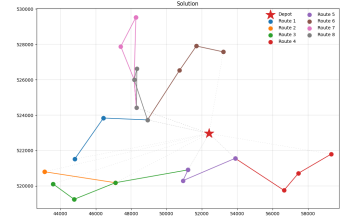




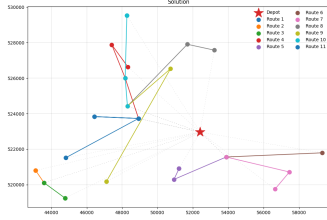
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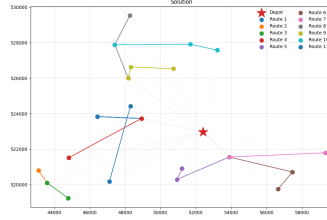
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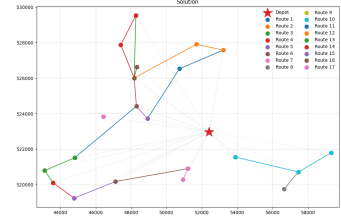
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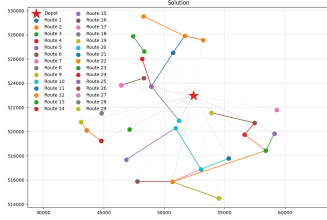


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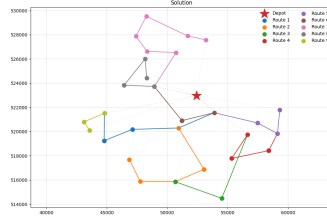


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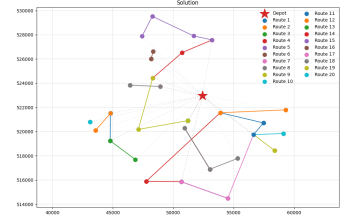
Figure 12: Routes for scenario 5 (K by FL).



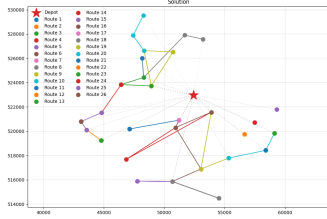
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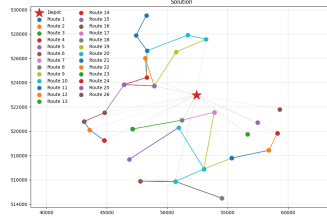
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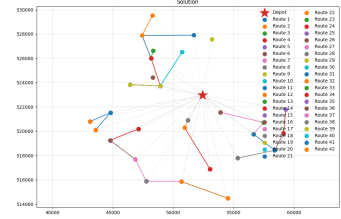
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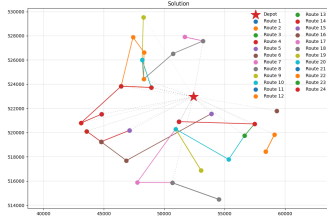


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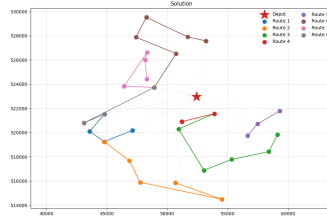


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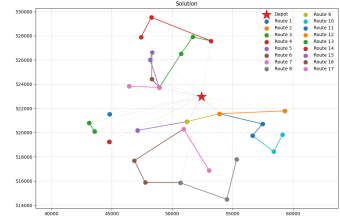
Figure 13: Routes for scenario 6 (J, AH).



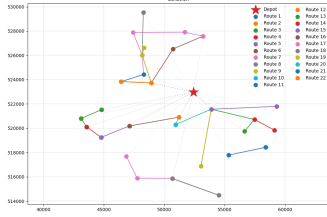
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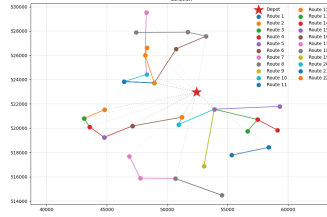
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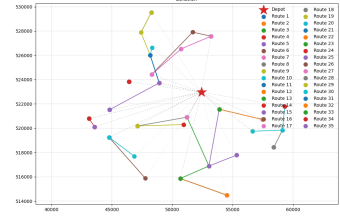
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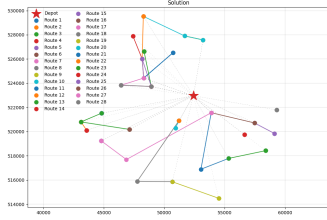


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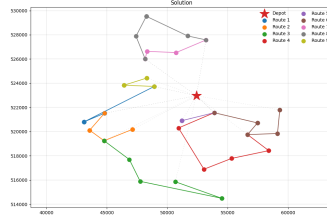


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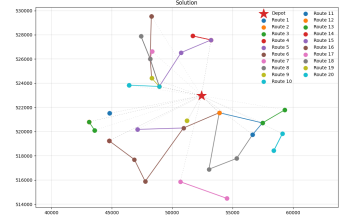
Figure 14: Routes for scenario 7 (J, K by FL).



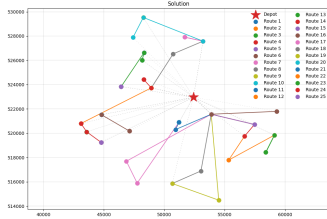
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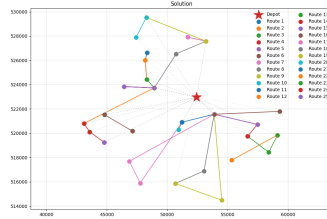
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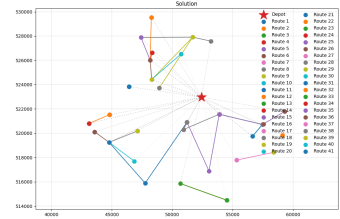
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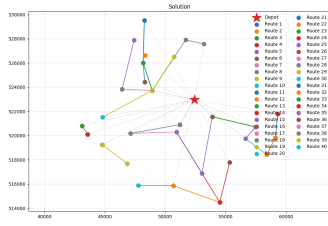


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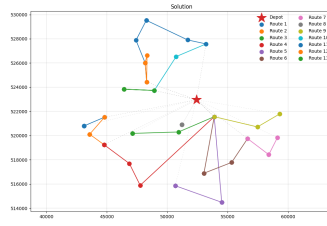


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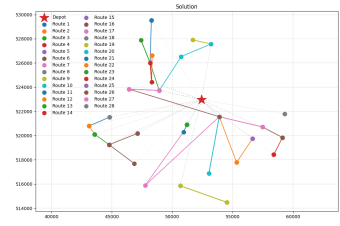
Figure 15: Routes for scenario 8 (AH, K by FL).



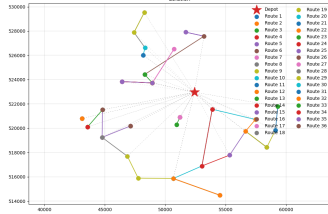
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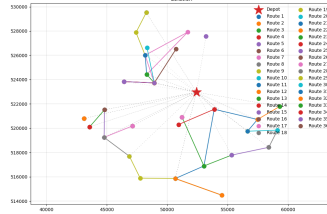
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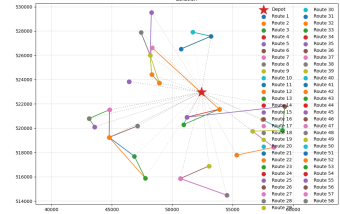
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(e) Friday



(f) Saturday

Figure 16: Routes for scenario 9 (J, AH, K by FL).